ML meets causal inference

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1 Augmented Inverse Propensity Weighting (AIPW)

1.1 Brief recap of IPW

- Under the previous assumption $\{Y_i(1) - Y_i(0)\} \perp T_i | X_i$ implying $\{Y_i(1) - Y_i(0)\} \perp T_i | p(X_i)$
- The ATE can also be consistently estimated as

$$\text{ATE} = \mathbb{E}[Y_i(1) - Y_i(0)] = \mathbb{E}\left[\frac{T_i}{p(X_i)} Y_i - \frac{(1 - T_i)}{1 - p(X_i)}\right] = \mathbb{E}\left[\frac{T_i - p(X_i)}{p(X_i)(1 - p(X_i))}\right]$$

- Thus, given the estimated propensity scores $\hat{p}(X_i)$ for each individual, we can construct the so-called inverse propensity weighted (IPW) estimator $\hat{\tau}_{\text{IPW}}$

$$\hat{\tau}_{\text{IPW}} = \frac{1}{N} \sum_{i=1}^{N} \left[Y_i \left(\frac{T_i}{\hat{p}(X_i)} - \frac{(1 - T_i)}{1 - \hat{p}(X_i)} \right) \right] = \frac{1}{N} \sum_{i=1}^{N} \left[Y_i \frac{T_i - \hat{p}(X_i)}{\hat{p}(X_i)\big(1 - \hat{p}(X_i)\big)} \right]$$

  - Weights up (or down) the over-represented (or under-) observations

1.1.1 Some issues

- Core issue with matching estimators remains: Requires common support/overlap
  - $0 < \mathbb{P}[T_i = 1 | X_i] < 1$
  - Overlap needs to be evaluated before applying the method by looking at the propensity score distribution
- When overlap is weak, models rely on extrapolation in regions where data is sparse
  - e.g., a single or a few observations get very large weights
  - **Extrapolation bias**
  - Often suggested to truncate propensity scores (e.g., min 0.01 and max 0.99)
- Large variance: The estimator typically has quite large variance and is not very stable

1.2 The key idea and equations

1.2.1 Doubly robust estimators – The AIPW

- Combines propensity-score based and regression-based estimation
- Estimators that only need the propensity score <u>or</u> the regression function to be correctly specified: The augmented IPW estimator (AIPW)

$$\text{ATE} = \frac{1}{N} \sum_{i=1}^{N} \left(\frac{T_i}{\hat{p}(X_i)} \big(Y_i - \hat{\mu}_{T=1}(X_i) + \hat{\mu}_{T=1}(X_i)\big) \right) - \frac{1}{N} \sum_{i=1}^{N} \left(\frac{(1 - T_i)}{1 - \hat{p}(X_i)} \big(Y_i - \hat{\mu}_{T=0}(X_i) + \hat{\mu}_{T=0}(X_i)\big) \right)$$

  - With $\hat{\mu}_{T=1}(X_i)$ and $\hat{\mu}_{T=0}(X_i)$ being the predicted value from a linear regression of X on Y for only treated or untreated observations
- Typical way to estimate standard errors / confidence intervals: bootstrap

- To better understand: rewrite in two parts:

$$\text{ATE} = \underbrace{\frac{1}{N}\sum_{i=1}^{N}\left(\hat{\mu}_{T=1}(X_i) - \hat{\mu}_{T=0}(X_i)\right)}_{\text{Simple regression adjustment}} + \underbrace{\frac{1}{N}\sum_{i=1}^{N}\left(\frac{T_i}{\hat{p}(X_i)}\left(Y_i - \hat{\mu}_{T=1}(X_i)\right) - \frac{(1-T_i)}{1-\hat{p}(X_i)}\left(Y_i - \hat{\mu}_{T=0}(X_i)\right)\right)}_{\text{IPW estimator on the residuals of the regression model } Y_i - \hat{\mu}(X_i)}$$

- If the regression model is correct, then $\mathbb{E}[Y_i - \hat{\mu}(X_i)|X_i]$ will be zero and the second part vanishes (i.e., we don't need the second part, because the first is already giving the right estimate)
- If the regression model is not correct, but the propensity score model is, the second part will properly correct (debias) the regression model estimates using the residuals
- However, $\hat{\mu}_{T=1}(X_i)$ and $\hat{\mu}_{T=0}(X_i)$ must not come from an OLS regression

1.3 AIPW and ML

- Two key components:
  - Outcome model:

    The predictions of Y given X (i.e., $\mathbb{E}[Y_i(1)|X_i] = \hat{\mu}_{T=1}(X_i)$ and $\mathbb{E}[Y_i(0)|X_i] = \hat{\mu}_{T=0}(X_i)$)
  - Propensity score model:

    The predictions of T given X ($\mathbb{P}[T_i = 1|X_i] = \hat{p}(X_i)$)
- Both can be any type of ML model that provides good predictions and may allow complex non-linearities and interactions among the X
  - $\hat{\mu}_{T=1}(X_i)$ and $\hat{\mu}_{T=0}(X_i)$ must not come from an OLS regression
  - The propensity score $\hat{p}(X_i)$ must come from logistic regression

1.4 Characteristics of the AIPW model

- Model misspecification:
  - The estimator is doubly robust: it requires only one of the two models – propensity score model or outcome model to be correct
  - However, if both are wrong, the bias can be larger than in any of the two separately
- Variance and instability in small samples
  - Usually, AIPW has smaller variance than the IPW estimator
    * AIPW achieves a semiparametric efficiency bound, meaninng that it attains the lowest possible asymptotic variance among consistent estimators when one of the two models is correct
    * IPW, even with a correct propensity model, does not achieve this efficiency
  - If the outcome model is overfitted, it can lead to poor generalization and erratic AIPW estimates, especially in small samples
    * Requires regularization and cross-fitting

2 Two key challenges with ML in causal inference

2.1 Regularization bias

- Regularization techniques (e.g., LASSO, ridge, elastic net etc.) deliberately shrink parameter estimates toward zero (or to another parameter) to reduce variance and prevent overfitting
- However, this shrinkage moves parameters away from their true estimates
  - They are no longer unbiased estimates of the true parameter
- Good for preduction because it tries to more strongly reduce variance than bias, but bad for causal inference
  - Therefore, using ML methods directly is usually not an option for causal inference
  - Double robustness (e.g., the AIPW) or orthogonalization (e.g., double ML) solves this problem

2.2 Overfitting bias

- The more flexible the ML, the more it tends to overfit
  - The model treats random fluctuations as significant patterns
  - Predictions contain traces of these random noise patterns
- When we use these predictions in our final causal equation (like the AIPW estimator or the double ML), the prediction errors of the ML models become correlated with the random errors of the outcome
  - They are not converging out to zero in expectation anymore
- This failure to average out creates a systematic overfitting bias
- How to solve this?
  - Predictions need to be independent of estimation / training $\rightarrow$ cross-fitting

3 Double (debiased) ML

3.1 The problem setting

- Can we use ML to more flexibly estimate the effects of covariates / control variables X_i
  - We still assume that we have conditional independence given X_i
  - We have a potentially high-dimensional X_i
- Problem: OLS regression can only take care of linear dependence of X_i
- We assume some more general non-linear dependence of X_i:

$$Y_i = \beta T_i + g(X_i) + \epsilon_i$$
$$T_i = m(X_i) + \xi_i$$

  - with $g(\cdot)$ and $m(\cdot)$ being any type of non-linear function that relates X_i with Y_i and T_i

3.2 Double ML (DML)

- We may want to use more flexible ML to estimate $\hat{g}(\cdot)$ and $\hat{m}(\cdot)$
  - Modern flexible ML are good in addressing non-linearities
  - Modern flexible ML may be well regularized and perform (implicit) variable selection
- Problem:
  - ML focuses on prediction and not on unbiased estimation of single parameters; thus we want to utilize the predictive ability of ML (and not the parameter estimates)
    - We don't need an interpretable ML; we will only need the predictions
    - At the end, any form of ML can be used, such as a simple lasso, random forest, LGBM, NNs etc
  - Regularization bias
    - Is addressed by double orthogonalization, utilizing the Frish-Waugh-Lowell theorem
  - Overfitting bias
    - Is addressed using sample splitting or cross-splitting

3.3 Frish-Waugh-Lowell theorem / Orthogonalization

- In a simple form, the theorem states that, when estimating a model on the form

$$Y_i = \alpha + \beta_1 T_i + \beta_2 X_i + \epsilon_i$$

- Then, the following estimators of β_1 are equivalent:
  - The OLS estimator obtained by regressing Y on T and X
    - i.e., the usual full regression approach
  - The OLS estimator obtained by regressing Y on $\tilde{T}$
    - Where $\tilde{T}$ is the residual from the regression of T on X (i.e., $\tilde{T} = T - \hat{\delta}X$)
    - The idea is that we are isolating the variation in T that is orthogonal (uncorrelated) to X
  - The OLS estimator obtained by regressing $\tilde{Y}$ on $\tilde{T}$
    - Where $\tilde{Y}$ is the residual from the regression of Y on X (i.e., $\tilde{Y} = Y - \hat{\gamma}X$)

3.4 Key concept of double ML

- A more general formulation of the same idea is the following, which does not require OLS estimates:

$$(Y_i - \mathbb{E}[Y_i|X_i]) = \beta(T_i - \mathbb{E}[T_i|X_i]) + \epsilon_i$$

- Thus, we can calculate orthogonalized regressors using more flexible non-linear functions $\hat{g}(\cdot)$ and $\hat{m}(\cdot)$
  - $\tilde{T}_i = T_i - \hat{m}(X_i)$
  - $\tilde{Y}_i = Y_i - \hat{g}(X_i)$
- We can estimate $\tilde{Y}_i = \beta\tilde{T}_i + \xi_i$ to receive a treatment effect estimate β for T
  - This reduces a high-dimensional multicariate and non-linear model into a simple bivariate regression

3.5 Double ML

- We use so-called orthogonal moment conditions (because both residuals have partialed out of the contribution of X, the only variation left in Y and T is what is orthogonal to X)
- Separates the estimation of the focal (causal) parameter from the estimation of the covariate model
  - We can use a simple linear and parametric model for focal effect
  - We can use a flexible non-linear model for the covariates without the need to make parametric assumptions
  - The coefficients on $\tilde{T}$ are uncorrelated with the estimation errors in $\hat{g}(X_i)$ or $\hat{m}(X_i)$
    - Prevents small errors in the estimation of g and m to propagate into the final target parameter
    - However, not as strong as double-robustness (requires both models to be approximately correct)
  - This is also similar to the way post-double selection avoids regularization bias

3.6 Sample splitting and cross-fitting

- We do this to avoid overfitting bias
- Sample splitting: train ML models on one part of the data and estimate treatment on the other half
- Cross-fitting: Reverse sample splitting and average estimates $\beta = \frac{\beta_1+\beta_2}{2}$
- We can also split into more than two groups (typically 5; similar to k-fold CV)

3.7 Double ML vs. AIPW

AIPW

- Allows only binary treatment variable T
  - It requires the propensity score $p(X_i)$
- Instability of division
  - AIPW involves IPW and thus must divide by the propensity score. When ps are close to zero we get very large weights, leading to massive variance and unstable estimates
- Focuses only on the ATE
  - In fact, AIPW can be viewed as a special case of Double ML

Double ML

- Allows binary or continuous treatment variable
  - It uses the expected value of the treatment, which is more general
- Mathematically more stable
  - Partialling-out approach (residualizing) does not require division by near zero even when treatment assignment is highly predictable
- Different types of treatment effect estimates
  - Allows for heterogenous effects (CATE)
  - Allows for instrumental variables (LATE)
  - ...

3.8 Partial linear vs. interactive model

- The model we have considered so far is called the partially linear treatment effect model, or partial linear regression (PLR) model
  - $Y_i = \beta T_i + g(X_i) + \epsilon_i$
  - The treatment effect is linear, and the rest can be non-linear
  - No treatment heterogeneity: the effect β is constant
- Interactive model / heterogenous treatment effect model
  - $Y_i = g(T_i, X_i) + \epsilon_i$
  - Allows for interaction of the treatment effect with all covariates (i.e., heterogenous effects)
  - However, focuses on estimating an overall effect (ATE or ATET) and not individual CATE
  - Typically, requires binary treatment $D_i \in \{0, 1\}$

3.9 Interactive model / heterogenous treatment effect model

- Estimate different outcome models $\hat{g}$ for $T = 1$ and $T = 0$ (using cross-fitting)

$$\hat{g}(1, X_i), \hat{g}(0, X_i), \hat{m}(X_i)$$

- Calculate orthogonalized score

$$\hat{\phi}_i = \hat{g}(1, X_i) - \hat{g}(0, X_i) + \frac{T_i\big(Y_i - \hat{g}(1, X_i)\big)}{\hat{m}(X_i)} - \frac{(1 - T_i)\big(Y_i - \hat{g}(0, X_i)\big)}{1 - \hat{m}(X_i)}$$

- Calculate ATE as

$$\hat{\theta} = \frac{1}{n} \sum_{i=1}^{n} \hat{\phi}_i$$

3.10 Note of caution

- We still need to have all relevant condounders in the initial set of X_i
  - We assume conditional independence
  - We only achieve a more flexible model (non-linear, interactions etc.)
- We do not have any colliders in the initial set of X_i

4 A brief outlook on causal trees

Key idea:

- Main objective: Estimate heterogenous treatment effects (CATE) rather than simply predicting outcomes
- Splitting logic: Unlike standard trees that split to minimize prediction error, causal trees split the data to maximize the difference in treatment effects between subgroups
- Cross-fitting / "honest trees": Prevent overfitting by splitting the training data in half: One half determines the tree structure (the splits), and the other half estimates the causal effect within those leavevs
- Compared to DML
  - Double ML focuses on high-dimensional confounders to estimate an unbiased overall effect (ATE)
  - Causal trees focus on discovering where and how that effect varies across different subgroups, generating interpretable, rule-based segments (e.g., if age more than 40, effect = X)
  - Combinations of both ideas exist: build the tree on the orthogonalized residuals (AIPW scores)